PORTFOLIO — SELECTED WORK

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A running catalog of things I've designed and built across mechanical engineering and EECS — from embedded firmware and circuit-level simulation to CAD-driven fabrication and applied mathematical modeling. Each project below includes the approach, the specs, and where to find the source or write-up.

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Interactive versions of every project below — exploded-view CAD viewers, live embeds, full photo galleries — are on the site at the link above.

CONTENTS

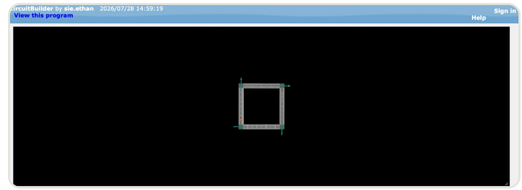
SPICE Solver for Resistor & EMF Networks

2025 `software`

Low-level SPICE-style solver for arbitrary resistor/EMF circuit networks.

A from-scratch circuit solver built in VPython: given an arbitrary network of resistors and EMF sources, it assembles Kirchhoff's Voltage and Current Laws together with Ohm's Law into a linear system and solves it directly, returning the current and voltage on every wire in the circuit.

The whole network — resistor values, EMF values, and the number/topology of circuits — is configurable, so it works as a general-purpose low-level SPICE-style solver rather than a fixed demo circuit.



METHOD	KVL + KCL + Ohm's Law, solved via linear algebra
PLATFORM	VPython (GlowScript) + JavaScript
OUTPUT	Current and voltage per wire
CONFIGURABLE	Resistor values, EMF values, circuit topology

IMU-Based Driver Monitoring System

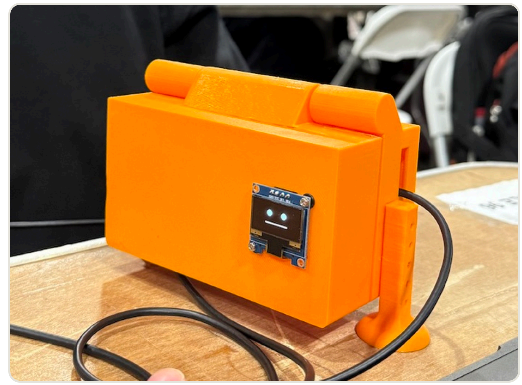
2026 `firmware` `software`

Camera-free driver-coaching wearable built in 36 hours at Purdue's StarkHacks, using dual ESP32-S3 IMUs and real-time audio feedback.

A non-invasive driver-monitoring co-pilot built in 36 hours at Purdue University's StarkHacks — one of the world's largest hardware hackathons — with a team of four. Instead of cameras or computer vision, a dashboard module and a wearable earpiece each carry an ESP32-S3 and an MPU-6050 IMU, fusing accelerometer/gyroscope data on-device to detect hard braking, sharp turns, aggressive acceleration, and lane changes, then relay coaching cues through audio alone so the driver never has to look away from the road.

I worked the embedded side: configuring the dual ESP32-S3s and IMUs in Arduino/C++, the ESP-NOW link synchronizing the two devices, and the audio delivery pipeline — a WebSocket server hosted directly on the ESP32's Wi-Fi, serving ElevenLabs-generated voice lines out of LittleFS and using the Web Audio API for stereo panning in the browser. An animated OLED face on the dashboard keeps the system feeling approachable rather than clinical.

Devpost — `https://devpost.com/software/passenger-princess` · GitHub — `https://github.com/ethannsie/Amoeba-StarkHacks`



HARDWARE	Dual ESP32-S3 + MPU-6050 IMU (dashboard + earpiece)
SYNC	ESP-NOW inter-device link
AUDIO PIPELINE	WebSocket server on ESP32 Wi-Fi → LittleFS → ElevenLabs voice lines → Web Audio API
DISPLAY	0.96" OLED, animated face UI
DETECTION	Hard braking, sharp turns, aggressive acceleration, lane changes
EVENT	Purdue StarkHacks — 36-hour hackathon, team of 4

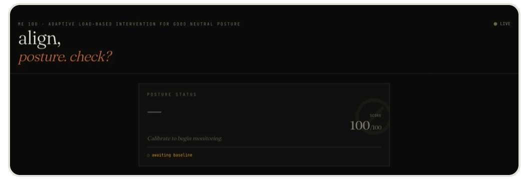
Embedded Posture Training System

2026 `firmware` `software`

Wearable dual-ESP32 posture coach that detects upper-body deviations and delivers real-time corrective feedback.

A real-time wearable posture-training system, built at UC Berkeley, using dual ESP32-S3 microcontrollers and IMUs to detect upper-body posture deviations and deliver immediate corrective feedback through non-blocking actuator patterns.

The architecture is distributed: sensing nodes read orientation data and publish posture states over MQTT, while a separate actuator node handles graduated feedback responses with its own timing so corrections stay responsive without blocking the sensing loop. Wi-Fi communication, JSON serialization, and state-based logic tie the loop together into a low-latency embedded feedback system, exploring how subtle, non-disruptive feedback can build long-term posture awareness.



HARDWARE	Dual ESP32-S3 + IMU sensing nodes + actuator node
SENSING	Upper (T4–T6) and lower (L3–L5) orientation tracking
MESSAGING	MQTT over Wi-Fi, JSON-serialized posture states
FEEDBACK	Non-blocking, graduated actuator response
CONTEXT	University of California, Berkeley

Picking the Perfect Pet

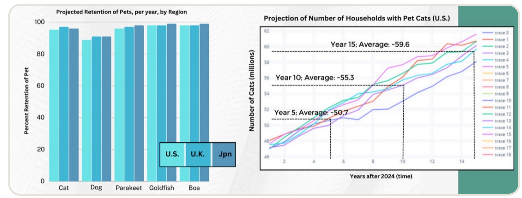
2024 `math`

A household readiness model for pet ownership that promotes adoption and cuts abandonment.

A household pet-readiness model built for the 2024 International Mathematical Modeling Challenge (Team US-14839), centered on HR-SPLIT — Hyper-Rectangular Space Partitioning with Linear Inequalities Technique — a method we developed to score a household's readiness for a pet from ten or fewer inputs (living space, free time, budget, species-specific care knowledge) without collapsing everything to a binary yes/no.

Fit the model to national data for the U.S., U.K., and Japan using normal and gamma distributions, then generalized it from cats to four more species — dogs, parakeets, goldfish, and red-tailed boas — and projected household counts and pet-retention rates 5, 10, and 15 years out using stochastic differential equations (Euler–Maruyama) and lifetime density functions.

Read the paper — `https://ethansie.vercel.app/assets/papers/immc-2024-picking-the-perfect-pet.pdf`



METHOD	HR-SPLIT — custom hyper-rectangular space partitioning + linear inequalities
SPECIES MODELED	Cat, dog, parakeet, goldfish, red-tailed boa
REGIONS	United States, United Kingdom, Japan
FORECASTING	Stochastic differential equations (Euler–Maruyama), lifetime density functions
KEY RESULT	29.2% of U.S. households ready for a cat; 95.3% cat retention vs. 88.8% dog retention

Global Sports League Scheduling

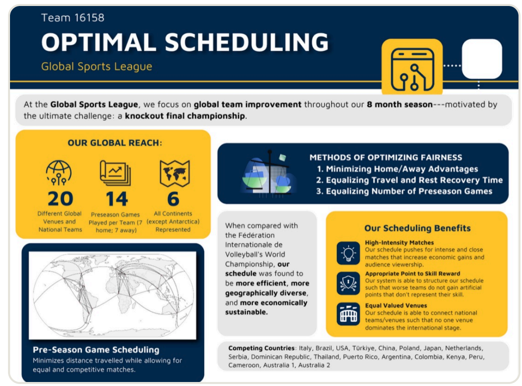
2025 math

A fairness- and travel-optimized scheduling system for a 20-team global volleyball league.

A scheduling system for a hypothetical 20-team Global Sports League (Team US-16158), commissioned to balance fairness, travel sustainability, and competitive equity across continents. We picked women's volleyball using a CEEG framework (Competitive equity, Economic sustainability, Environmental sustainability, Growth potential) built on Olympic-era medal data, then selected 20 founding teams spanning six continents with a FIVB-based strength-rating system.

Match-ups were generated with a mixed-integer linear program plus a Markov-chain analysis to balance total travel distance against competitive fairness, and a second MILP scheduled those matches across an 8-month season while respecting rest days and home/away balance. We scored the resulting schedule for 'effectivity' (does the final standing reflect true team skill) and 'attractiveness' (closeness, intensity, entropy) via Monte Carlo simulation, and benchmarked it against the real FIVB Women's World Championship schedule.

Read the paper — https://ethansie.vercel.app/assets/papers/immc-2025-global-sports-league.pdf



METHOD	Mixed-integer linear programming (PuLP) + Markov chain analysis
SPORT SELECTED	Women's volleyball, via the CEEG selection framework
SEASON	20 teams (6 continents), expanded to 24; 8-month season
EVALUATION	Monte Carlo simulation of effectivity & attractiveness
TOOLS	Python — PuLP, NetworkX, GeoPandas, SciPy
RECOGNITION	Finalist, 2025 IM²C USA Regional Contest

Award Manufacturing [Berkeley Engineering Youth Academy]

2026 mechanical

A project focused on the design and fabrication of custom awards for the Berkeley Engineering Youth Academy (BEYA).

A project focused on the design and fabrication of custom awards for the Berkeley Engineering Youth Academy (BEYA). The awards were created using a combination of 3D printing and traditional manufacturing techniques, allowing for a high degree of customization and personalization.

The project involved collaboration with BEYA staff and students to ensure the awards met their needs and expectations. Feedback was gathered throughout the design process, leading to several iterations and refinements.

METHOD	3D printing + traditional manufacturing
MATERIALS	PLA filament, cast acrylic sheet, metal hardware
TOOLS	Fusion 360, Adobe Illustrator, Bambu Labs Slicer, Bambu Labs X1 Carbon
RECOGNITION	Featured in BEYA 2026 showcase

Olympic Medal Tables

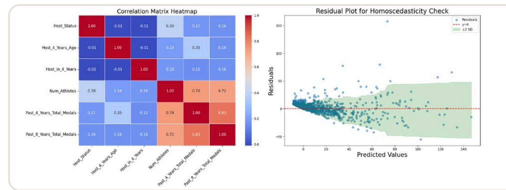
2025 math

A hierarchical statistical model forecasting the 2028 LA Olympics medal table from 128 years of IOC data.

A hierarchical statistical model (Team 2522820, MCM/ICM Problem C) forecasting the medal table for the 2028 Los Angeles Summer Olympics from 128 years of IOC data. Started from a Tobit model — handling the fact that medal counts are zero-inflated, censored data — fit by maximum likelihood, then found it suffered from heteroscedasticity and moved to a hierarchical mixed-effects framework: countries were clustered by historical performance with DBSCAN, then modeled with Negative Binomial regression plus a Probit 'hurdle' stage for countries likely to earn zero medals.

Beyond the medal forecast, used Singular Value Decomposition to rank which individual events most influence a country's ranking, and built a rolling-window time-series z-score model to detect the 'great coach effect' — anomalous, sustained performance jumps for a country in a given sport — cross-referencing the top hits against real historical Olympic coaches to sanity-check the model.

Read the paper — https://ethansie.vercel.app/assets/papers/mcm-2025-olympic-medal-tables.pdf



METHOD	Tobit model (MLE) → hierarchical mixed-effects Negative Binomial + Probit hurdle
CLUSTERING	DBSCAN on historical country performance
DATA	IOC medal & athlete data, 1896–2024, processed via SQLite
EXTENSIONS	SVD event-impact ranking; time-series 'great coach effect' detection
LANGUAGE	Python 3
HEADLINE PREDICTION	USA 48 golds, Germany 28, China 25 for LA 2028